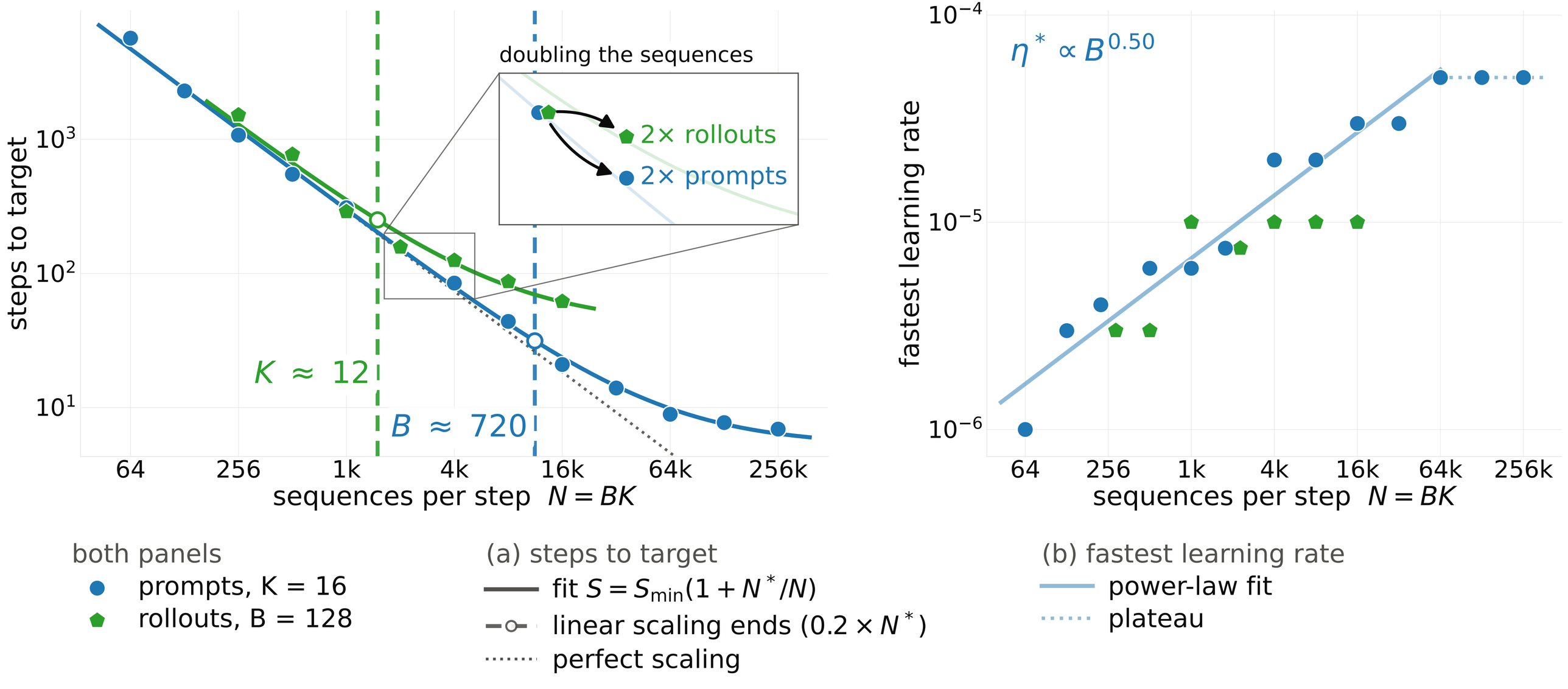
doubling the sequences
2× rollouts
2× prompts
$K \approx 12$
$B \approx 720$
steps to target
10^3
10^2
10^1
64
256
1k
4k
16k
64k
256k
sequences per step $N = BK$
$\eta^* \propto B^{0.50}$
fastest learning rate
10^{-4}
10^{-5}
10^{-6}
64
256
1k
4k
16k
64k
256k
sequences per step $N = BK$
both panels
prompts, K = 16
rollouts, B = 128
(a) steps to target
fit $S = S_{\min}(1 + N^*/N)$
linear scaling ends ($0.2 \times N^*$)
perfect scaling
(b) fastest learning rate
power-law fit
plateau